

YUANKAI TAO

Email: yunkaitao0909@gmail.com | Tel: +1 470 658 3699 | [GitHub](#) | [Homepage](#)

EDUCATION

Georgia Institute of Technology Aug 2026 — Dec 2027

M.S. in Operations Research, Stewart School of Industrial and Systems Engineering

Selected Coursework: Deterministic Optimization, Stochastic Optimization, Simulation

University of California, San Diego Sep 2024 — Dec 2025

M.S. in Quantitative Finance, Rady School of Management

GPA: 3.84/4.0

Selected Coursework: Advanced Risk Management, Computational Finance Methods, Data Science for Finance

Zhongnan University of Economics and Law

Sep 2020 — Jun 2024

B.S. in Management, School of Accounting and Finance

GPA: 3.63/4.0

Selected Coursework: Probability & Statistics, Linear Algebra, Calculus, Python Programming

TECHNICAL SKILLS

Programming & Tools

Python, SQL, MATLAB, MS Office, L^AT_EX, Git, Linux, GCP

Quantitative Methods

Advanced Risk Management (VAR, GARCH, Greeks, Delta Hedging),
Regression Analysis, Probability Theory, Time Series Analysis, Simulation

Libraries & Frameworks

Pandas, Polars, Numpy, Scipy, PyTorch, scikit-learn, CVXPY, Backtrader

PROFESSIONAL EXPERIENCE

Guangdong Hedge Fund Investment Management Co., Ltd.

May 2026 — Jul 2026

Quantitative Research, Intern

Quantitative Strategy Department

- Built a dynamic **VaR/CVaR risk measurement and model evaluation framework** for copper futures, comparing 17 models across five volatility specifications (including GARCH and EWMA) and Normal, Student-t, Skewed-t, and Filtered Historical Simulation assumptions; used semi-variance slicing to identify long/short momentum risk exposures.
- Engineered an end-to-end commodity futures **statistical and machine learning modeling** pipeline with 16 trend, momentum, volatility, liquidity, volume, and open-interest features; trained and ensemble LightGBM, XGBoost, CatBoost, and neural networks across 1- to 20-day horizons.
- Applied **rolling out-of-sample model validation** using statistical and trading metrics; implemented a monthly rebalanced strategy with 120-day rolling retraining that achieved a **1.89 Sharpe ratio, 16.79% annualized return**, and **25.51% excess return** after transaction costs.

VariFlight Technology

Jul 2025 — Sep 2025

Data Analyst, Intern

Data Development Department

- Developed Python web-scraping workflows to collect public reports, webpages, and images from the Civil Aviation Administration of China, supporting structured **data management** and downstream analytical use.
- Applied OCR and table-extraction tools, including pyesseract and Camelot, to convert **600+ unstructured PDF reports and 500+ webpages** into standardized CSV datasets for local LLM training, improving data quality and repeatability.

Ernst & Young

Dec 2022 — Feb 2023

Assurance, Intern

Financial Service Office (FSO)

- Supported the audit of intercompany and investment transactions for Hyundai Motor Finance Company; performed **variance analysis, risk assessment, and exception investigation** to identify and document significant deviations.

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel HillMar 2025 — Jan 2026

The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining

- Engineered a **large-scale data pipeline** (Python, PostgreSQL, PostGIS) for 76 million spatial observations; managed a **964.5M-record, 100GB+ SQL database** on Google Cloud Platform and UNC Longleaf HPC using Python and DuckDB.
- Designed parallelized spatial computation and property-specific data controls across multiple distance ranges, delivering approximately **50× computational speedup** for scalable analytical workflows.
- Executed complex geospatial queries with time-window and property-level constraints to produce reproducible population statistics for quantitative research.

SELECTED PROJECTS

UCSD / FMSbonds, Inc. 2025 Capstone ProjectAug 2025 — Dec 2025

Team Lead | High-Performance Bond Pricing Engine Optimization

- Developed a municipal bond **risk-screening and control framework** covering credit, structural, sector, tax, interest-rate, and data-integrity risks to support disciplined, data-driven risk identification.
- Designed caching for high-frequency cash-flow queries, eliminating redundant object generation and reducing per-bond solving time from 1.17 ms to 0.13 ms (**9× speedup**).
- Reimplemented the yield-to-worst numerical solver with Numba JIT, reducing runtime from 1.57 ms to 7.5 μs per bond (**210× speedup**) and enabling scalable portfolio analytics.

Kaggle Jane Street Market Prediction CompetitionSilver Medal

- Processed **47M minute-level observations** spanning 1,669 trading days; engineered cross-sectional, lagged, rolling, and market-regime features from 79 anonymized variables and 9 targets.
- Trained and ensembled LightGBM, CatBoost, XGBoost, MLP, and TabNet models with **time-aware validation**; placed **94th (Top 2.5%)** of 3,757 teams on the private leaderboard.

CERTIFICATIONS & HONORS

Deep Learning Specialization (5-Course Certificate)	Coursera, DeepLearning.AI
Algorithms and Data Structures Certificate (6-Course Certificate)	Coursera, UC San Diego
Beta Gamma Sigma (Top 20% of graduating class)	UCSD, 2024 — 2025
WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%)	2024
Special Scholarship in Innovation and Entrepreneurship	ZUEL, 2022 — 2023